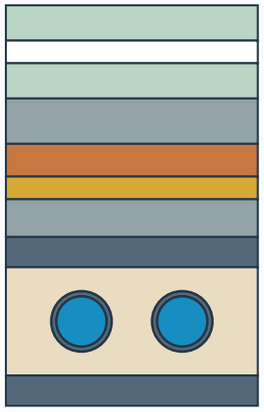
# Pixel barrel: material and mounting

Module normal points outward toward the stave. Views use independent display scales.

## A. Through the module and cooled stave

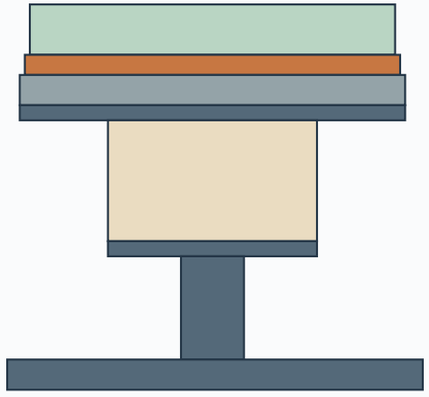


Sensor silicon: 0.150 mm
Bump standoff (unfilled): 0.020 mm
ASIC silicon: 0.150 mm
Graphite contact shim: 0.255 mm
Epoxy support interface: 0.100 mm
Electrical insulation: 0.025 mm
Graphite spreader: 0.200 mm
CFRP upper skin: 0.150 mm
Bored foam core: 4.100 mm
CFRP lower skin: 0.150 mm

**Support beyond body-back plane: 4.725 mm total.**

0.080 mm-equivalent core glue retains specified foam mass.
Flex-side epoxy (0.025 mm) is on the opposite sensor face.
Unknown graphite/ASIC and mount bondlines remain open.

## B. At one bearing station (radial section)



Module
Epoxy + insulation
Narrow spine
10 / 24 mm
CFRP foot
4 mm wide, 8 mm in z
Ring outside stave
Cage / end support

Dashed: joint/cage topology; thickness and stiffness unspecified.

## C. Axial load path and station schedule

Cage and end restraints: conceptual. Rings are coupled parts of one support, not six ground reactions.

z = -546 mm sets axial reference; the other contacts propose longitudinal sliding/compliance.

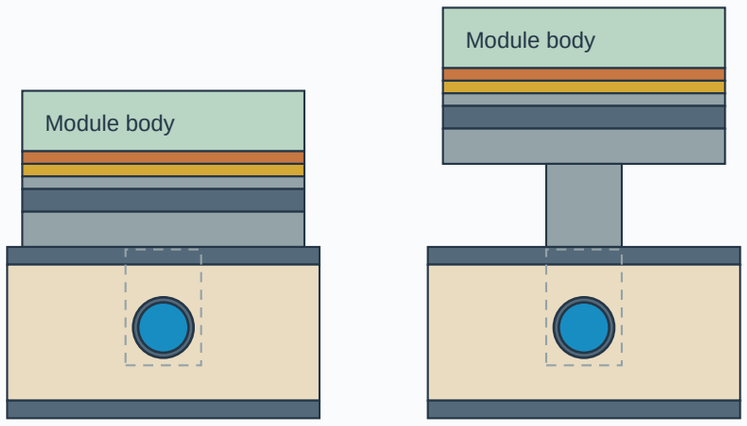
**PROTOTYPE / REVIEW: dimensions label the model; joints and stiffness remain unqualified.**

DES012 PB-C03-05; DES002 PS-C01-10/I04; tools/pixel_support/mounting.py make_mounts/check_mounts.

# Pixel endcap: graphite feet and tongues

Local per-chip thermal contact and global support are different interfaces.

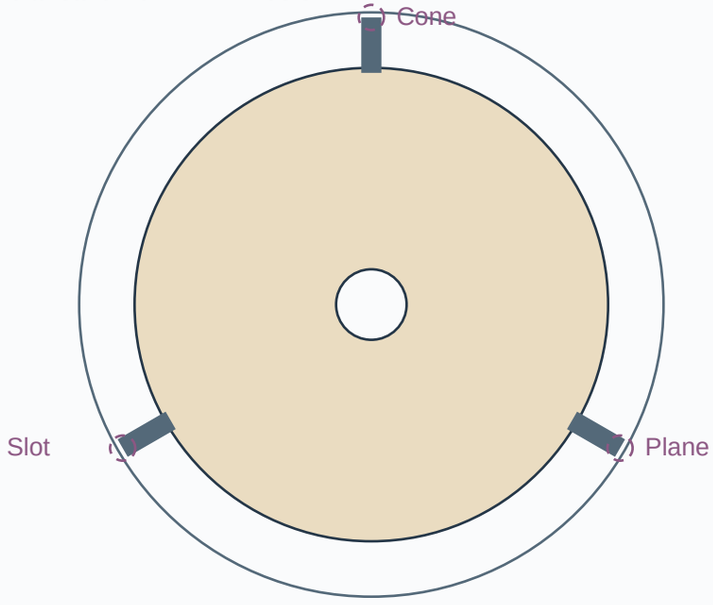
## A. Local graphite foot (separate illustrative cuts)




Pickup = 0.30 graphite + 0.075 CFRP cradle + 0.025 each
insulation, TIM and adhesive = 0.45 mm. Raised stem: 6 x 8 x 1.65 mm.
Dashed insert: effective core inventory; tube saddle/contact needs CAD.

## B. Three tongues in disc plan

Full annular sandwich r = 27..188.5 mm




Tongues: 10 mm wide, r = 186..228 mm; angles 90/210/330 deg.
Rails: 8 x 8 mm boxes, wall 0.4 mm, on closed CFRP shell.
Cone/slot/plane is a proposed 6-constraint kinematic coupling.

## C. Load transfer beyond the disc

Module cradle / graphite foot → Common disc sandwich → Three radial tongues → Three box rails + closed shell → End flanges / external support

Rail-to-shell bond, coupling preload, cold motion and end reactions require analysis and coupon/assembly tests.

Whole-endcap axial installation is assumed; shell split/access and independent half-disc extraction remain open.

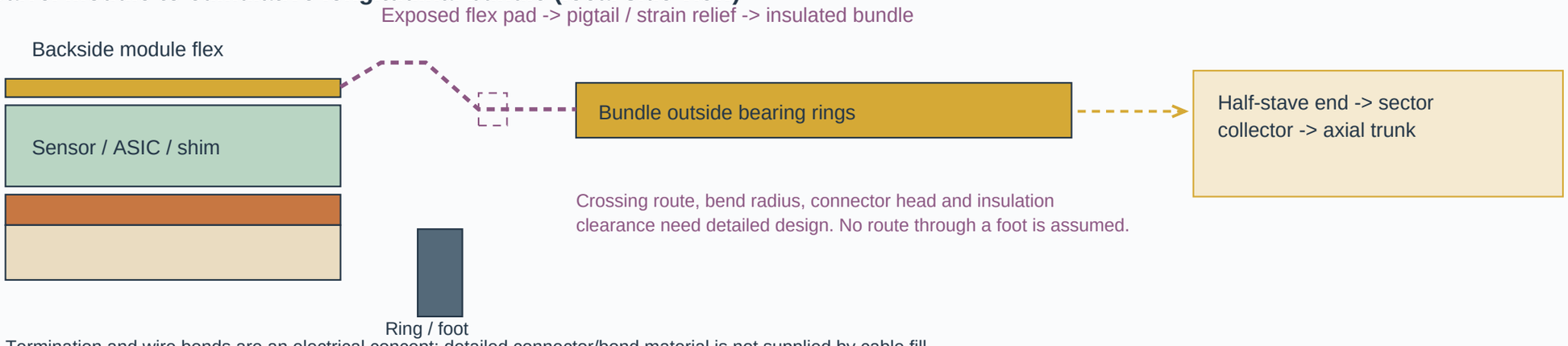
**PROTOTYPE / REVIEW: dimensions label the model; joints and stiffness remain unqualified.**

DES019 PM-C04-06; DES014 EC-C07/10/13-16; TDR current chapter support ledger. Exact input SHAs in README.

# Pixel services: proposed pigtail handoffs

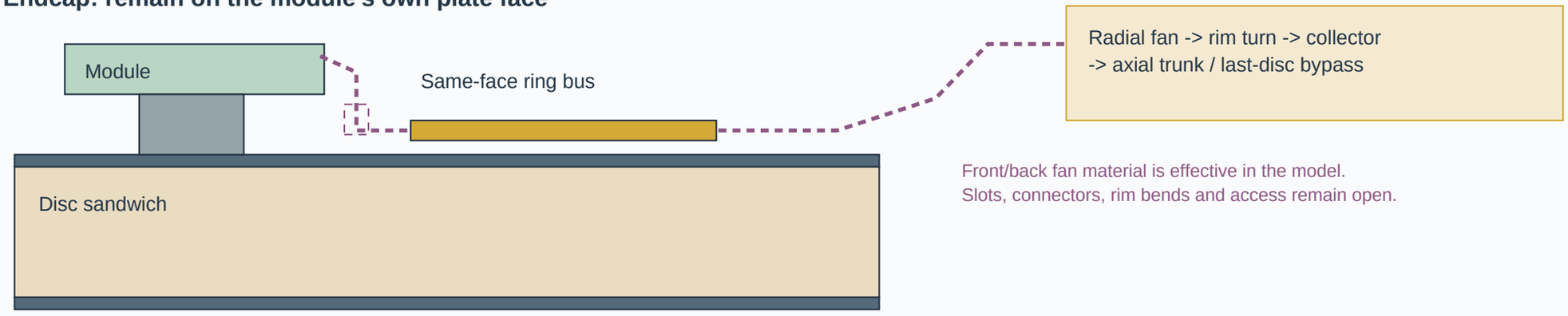
Solid blocks locate material/envelopes; dashed paths and open connectors are conceptual interfaces.

## A. Barrel module to cumulative longitudinal bundle (local side view)




Termination and wire bonds are an electrical concept; detailed connector/bond material is not supplied by cable fill.

## B. Endcap: remain on the module's own plate face




Side-slot drop beside the foot; no passage through an opposite-face module.

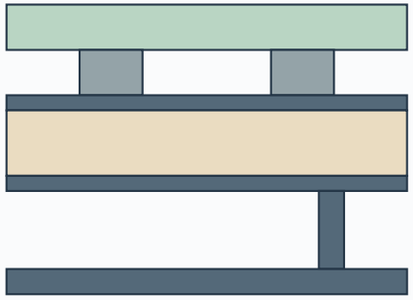
Electrical chains and hydraulic circuits are separate networks; a graphite foot is not a coolant feed or connector.

**PROTOTYPE / REVIEW: dimensions label the model; joints and stiffness remain unqualified.**

# Strip mounting: local load to external frame

One cooled structural sandwich per stave/petal; the long-strip pair uses both faces of that same carrier.

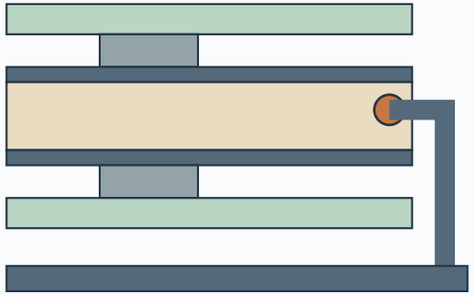
## A. Short-strip barrel



Module -> pickups ->
sandwich -> bare skin land
-> 2 mm beveled web
-> outer bearing ring

9 stations. Centre fixes z; others slide/comply.
Ring/cage reaction and web joint need qualification.

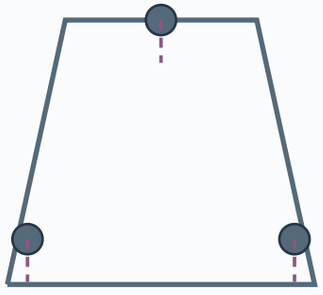
## B. Long-strip barrel



Both sensors -> same
sandwich -> potted lands
-> keyed web / shoe
-> inner bearing ring

17 stations, 163.75 mm pitch. Centre fixes z.
Other contacts reserve +/-0.3 mm longitudinal travel.

## C. Short-strip petal

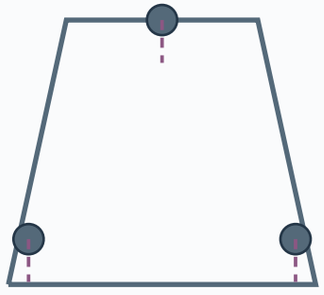


**Module pickups -> petal**

Three back lugs to inner/outer CFRP frames. Inner datum; two outer constrained/sliding seats. Frame-to-cage joint remains open.

Lug preload, cold adhesive stress and cage stiffness are open.

## D. Long-strip petal



**Both faces -> common petal**

Two outer contacts + inner key. One outer fixes in plane; other slides tangentially; inner slides radially. +/-0.3 mm reserve.

Horizontal handling fails screen; temporary frame required.

**PROTOTYPE / REVIEW: dimensions label the model; joints and stiffness remain unqualified.**

DES020 SB-C04/07/15; DES021 SE-C04/06; DES022 LS-C02-04; DES023 LE-C02/03/07; current TDR bearing schedule.